

Professional Git & GitHub Lab Assignment Report

Student Details

Name: _____
Roll No.: _____
Course: _____

Aim

Perform Git branching, merging, conflict resolution, reset, rebase, and GitHub collaboration using a Flask project.

Task 1

Create GitHub repository, generate SSH key if needed, clone repository, create username branch, add Flask project, commit, merge into main.

Commands

```
ssh-keygen -t ed25519 -C "email@example.com"  
git clone git@github.com:USERNAME/REPO.git  
git checkout -b USERNAME  
git add .  
git commit -m 'Add Flask project'  
git checkout main  
git merge USERNAME  
git push origin main
```

Screenshot: Paste terminal and GitHub branch screenshot here.

Task 2

Create branch USERNAME_new, update JSON for /api, commit, merge into main. If conflict occurs, keep USERNAME_new changes, then:

```
git add .  
git commit -m 'Resolve merge conflict'  
git push origin main
```

Screenshot: Paste merge/conflict screenshots.

Task 3

Create branches master_1 and master_2. In master_1 build To-Do form (Item Name, Item Description). In master_2 create POST /submittodoitem storing data in MongoDB Atlas. Merge both branches into main.

Screenshot: Paste frontend, API, MongoDB and merge screenshots.

Task 4

In master_1 add Item ID, UUID and Hash using three separate commits.

```
git add . && git commit -m 'Add Item ID'
```

```
git add . && git commit -m 'Add Item UUID'
git add . && git commit -m 'Add Item Hash'
Merge into main. Roll back to Item ID commit using:
git reset --soft <commit-id>
git commit -m 'Recommit Item ID state'
git rebase main master_1
Preserve commit history (do not squash).
```

Screenshot: Paste git log before/after reset and rebase.

GitHub Repository

Repository: <https://github.com/your-username/flask-git-assignment>

Conclusion

This assignment demonstrated GitHub collaboration, branching, merging, conflict resolution, Git reset, rebasing, and integration of Flask and MongoDB features using best practices.